Glossary of Terms

The number in brackets is the lecture where the term is explained

Absolute magnitude (M) the apparent magnitude a star would have at a distance of 10 parsecs; a measure of its true brightness. [3.1]

Adhik Maas the extra lunar month added about every 32½ months to keep the Hindu lunisolar calendar in step with the seasons. [1.4]

Altitude angle of an object above the horizon (0° at the horizon, 90° at the zenith). [2.1, 2.2]

Amanta / Purnimanta lunar months ending at New Moon / Full Moon. [1.4]

Amavasya New Moon; the 30th tithi. [1.3, 1.4]

Aperture (D) diameter of a telescope's main lens or mirror; its most important specification. [3.2]

Aphelion / Perihelion Earth's farthest point from the Sun (~4 July) / closest point (~3 January). [1.2]

Apparent field of view (AFOV) angular width of the view seen through an eyepiece alone (e.g. 52°). [3.3]

Apparent magnitude (m) how bright an object looks from Earth; smaller numbers are brighter. [3.1]

Asterism a recognisable star pattern that is not an official constellation (e.g. the Big Dipper). [2.4]

Astronomical unit (AU) average Earth–Sun distance, 149.6 million km. [1.2]

Averted vision looking slightly to one side of a faint object to see it better. [2.5]

Ayanamsha the angle ($\approx$ 24° today) between the tropical and sidereal zodiacs, caused by precession. [1.2]

Azimuth angle along the horizon measured from North towards East (N = 0°, E = 90°, S = 180°, W = 270°). [2.2]

Barlow lens a lens placed before the eyepiece to multiply magnification (usually 2×). [3.2]

Bayer designation a star's name using a Greek letter plus the constellation's genitive, e.g. α Orionis. [2.3]

Bortle scale a 1–9 scale of night-sky darkness (1 = darkest). [2.5]

Catadioptric telescope uses both mirrors and lenses (Schmidt-Cassegrain, Maksutov). [3.2]

Celestial equator Earth's equator projected onto the sky; Dec 0°. [2.1]

Celestial poles the points in the sky directly above Earth's poles, around which the sky rotates. [2.1]

Celestial sphere the imaginary sphere around Earth on which we map the positions of objects. [2.1]

Chromatic aberration coloured fringes produced by a lens because it bends different colours by different amounts. [3.2]

Circumpolar a star that never sets at your latitude ($\delta > 90° - \varphi$). [2.1]

Collimation alignment of a telescope's optics, especially the mirrors of a Newtonian. [3.4]

Colour index (B − V) difference between a star's blue and visual magnitudes; a measure of its colour and temperature. [3.1]

Constellation one of the 88 official regions of the sky defined by the IAU. [2.3]

Culmination (transit) the moment a star crosses the meridian; its highest point. [2.2]

Dark adaptation the eye's increase in sensitivity after 20–30 minutes in darkness. [2.5]

Dawes limit the smallest angular separation a telescope can resolve: $116/D$ arcsec (D in mm). [3.3]

Declination (δ, Dec) angle north (+) or south (−) of the celestial equator. [2.2]

Deep-sky object (DSO) clusters, nebulae and galaxies beyond the solar system. [3.5]

Distance modulus $m - M$; gives the distance via $d = 10^{(m-M+5)/5}$ pc. [3.1]

Diurnal motion the daily east-to-west movement of the sky, caused by Earth's rotation. [1.1]

Dobsonian a Newtonian telescope on a simple alt-azimuth box mount. [3.2]

Eclipse (solar / lunar) the Moon blocking the Sun (at New Moon) / the Moon entering Earth's shadow (at Full Moon). [1.3]

Ecliptic the Sun's apparent yearly path among the stars, tilted 23.44° to the celestial equator. [1.2]

Elongation the angle between an object (e.g. the Moon) and the Sun as seen from Earth. [1.3]

Epoch (J2000) the reference date for star coordinates, needed because of precession. [2.2]

Equation of time the difference between sundial time and clock (mean) time, up to ±16 minutes. [1.6]

Equatorial mount a mount with one axis parallel to Earth's axis, so stars can be followed by turning one knob. [3.2, 3.4]

Equinox when the Sun crosses the celestial equator (~20 March, ~23 September); day ≈ night everywhere. [1.2]

Exit pupil diameter of the beam of light leaving the eyepiece: D/M. [3.3]

Eyepiece the small lens system you look through; it sets the magnification. [3.2]

Flamsteed number a star number within a constellation, assigned in order of RA (e.g. 61 Cygni). [2.3]

Focal length (F) / focal ratio (N) distance from the objective to the focus / F ÷ D. [3.2, 3.3]

Ghati (ghatika) Indian time unit of 24 minutes (60 per day). [1.4]

Globular cluster an old, dense, spherical cluster of 100,000+ stars. [3.5]

Hijri calendar the Islamic lunar calendar counting from 622

CE. [1.5]

Hilal the thin crescent Moon whose sighting begins an Islamic month. [1.5]

Hora a planetary hour; the basis of the weekday order. [1.4]

Horizon the great circle 90° from the zenith. [2.1]

Hour angle (HA) LST − RA; time since a star crossed the meridian. [2.2]

IAU International Astronomical Union; it defines the constellations and official star names. [2.3]

IST Indian Standard Time, UTC + 5:30, based on 82.5° E. [1.6]

Leap year a year with 366 days (Gregorian rule: ÷4, except centuries not ÷400). [1.6]

Light-gathering power proportional to aperture area (D^2). [3.3]

Light-year (ly) the distance light travels in a year, 9.46 trillion km. [3.1]

Limiting magnitude the faintest magnitude visible with a given eye, instrument and sky. [2.5, 3.3]

Local sidereal time (LST) the right ascension currently on your meridian. [2.2]

Luminosity the total power output of a star. [3.1]

Magnitude the astronomical brightness scale; 5 magnitudes = 100× in brightness. [3.1]

Meridian the great circle through North, zenith and South. [2.1]

Messier catalogue Charles Messier's list of 110 bright deep-sky objects (M1–M110). [3.5]

Muhurta Indian time unit of 48 minutes. [1.4]

Nadir the point directly below the observer. [2.1]

Nakshatra one of 27 lunar mansions, each 13°20′ wide. [1.4]

NGC New General Catalogue (Dreyer, 1888) of 7,840 deep-sky objects. [3.5]

Nodes (Rahu & Ketu) the two points where the Moon's orbit crosses the ecliptic; eclipses occur near them. [1.3]

Obliquity of the ecliptic Earth's axial tilt, 23.44°. [1.2]

Open cluster a young, loose group of stars born together (e.g. the Pleiades). [3.5]

Opposition when a planet is opposite the Sun in the sky: up all night, closest and brightest. [2.5]

Parsec (pc) the distance at which a star shows 1″ of parallax; 3.26 light-years. [3.1]

Paksha a lunar fortnight: Shukla (waxing) or Krishna (waning). [1.4]

Panchang the Indian almanac giving tithi, vara, nakshatra, yoga and karana. [1.4]

Planetary nebula the gas shell ejected by a dying Sun-like star. [3.5]

Polar alignment pointing an equatorial mount's polar axis at the celestial pole. [3.4]

Precession the slow wobble of Earth's axis (one cycle in ~25,800 years). [1.2]

Prime vertical the great circle through East, zenith and West. [2.1]

Purnima Full Moon; the 15th tithi. [1.3, 1.4]

Rashi one of the 12 zodiac signs (30° each) in the Indian sidereal zodiac. [1.4, 2.4]

Reflector / Refractor a telescope using a mirror / a lens as its objective. [3.2]

Right ascension (α, RA) celestial "longitude", measured eastward from the First Point of Aries, in hours. [2.2]

Sankranti the Sun's entry into a new rashi (e.g. Makar Sankranti, Mesha Sankranti). [1.2]

Saros the 18-year 11-day cycle after which eclipses repeat. [1.3]

Seeing steadiness of the atmosphere; poor seeing makes images shimmer. [3.3, 3.4]

Sidereal day / month / year rotation, orbit or revolution measured relative to the stars. [1.1, 1.3]

Solstice when the Sun reaches its farthest north (~21 June) or south (~21–22 December). [1.2]

Surface brightness brightness per unit area of an extended object. [3.5]

Synodic month New Moon to New Moon, 29.53 days. [1.3]

Tithi lunar day; the time for the Moon to gain 12° on the Sun. [1.4]

Transparency clarity of the air (haze, dust, humidity). [3.4]

Tropical year equinox to equinox, 365.2422 days; the year of the seasons. [1.2]

True field of view (TFOV) the actual patch of sky seen in the eyepiece: AFOV/M. [3.3]

Umbra / Penumbra the dark central shadow / the lighter outer shadow. [1.3]

Uttarayana / Dakshinayana the Sun's northward / southward half-year journey. [1.2]

Vara weekday. [1.4]

Zenith the point directly overhead. [2.1]

Zero Shadow Day the day the noon Sun is exactly overhead and vertical objects cast no shadow. [1.2]

Zodiac the band of constellations along the ecliptic. [1.2, 2.4]